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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

C PROJECT REPORT ON

ONLINE

SUBMITTED TO

The Technical Department

KG College of Arts and Science

SUBMITTED BY

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Academic Year: **2025 – 2026**

Semester: **I**

DECLARATION

We hereby declare that the project entitled “ONLINE QUIZ SYSTEM” submitted in partial fulfilment of the requirements for the course [**BSc.computer science]** is a **record of our original work** carried out under the guidance of **[Mr.Moses John Prakash]**, Technical Trainer, Technical Department, KGiSL MicroCollege.

We further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

We have acknowledged all sources of information used in this project wherever applicable.

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CERTIFICATE

This is to certify that the project entitled **"ONLINE QUIZ SYSTEM"** is a **bona fide work** carried out by the following students of **[1 C/2025]**, [**BSc. Computer science]**, **KG College**

of Arts & Science, in partial fulfilment of the requirements for the course **[Bsc.computer science]**.

This project has been completed under my supervision and guidance during the academic period **[November, 2025]**.

It is further certified that this project is the **original work** of the students and has not been previously submitted for the award of any degree, diploma, or certificate at this or any other institution.

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ABSTRACT

The project **"Online Quiz System"** is a simple console-based application developed using the C programming language. It aims to provide an interactive and engaging platform for users to test their knowledge through multiple-choice questions. This particular implementation focuses on a **Food Quiz**, designed to educate and entertain users with fun facts about food.

The program utilizes fundamental C programming concepts such as **input/output functions**, **conditional statements**, and **character handling** through the ctype.h library. The quiz presents a series of questions, each with four possible answers, and allows the user to select an option by entering a corresponding character. The system evaluates the user's answers in real-time, calculates the total score, and displays the final result at the end of the session.

This project demonstrates how basic C programming constructs can be effectively used to build a simple interactive application. It serves as an introductory project for beginners to understand user interaction, decision-making structures, and modular programming in C.

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1. INTRODUCTION

1.1 Objective

- The main objective of this project is to develop a simple and interactive Online Quiz System using the C programming language.
- The system aims to engage users by testing their knowledge on various topics—in this case, food—through a series of multiple-choice questions.
- It provides instant feedback through score calculation and helps users learn in a fun and educational way.

1.2 Overview

- The Online Quiz System is a console-based program designed to simulate a basic quiz environment.
- The quiz presents multiple-choice questions to the user, accepts their input, evaluates the

correctness of each answer, and calculates the total score.

- The program makes use of standard C libraries such as stdio.h for input and output operations and ctype.h for handling character inputs effectively.
- This simple structure demonstrates how conditional statements and user input can be combined to create an interactive learning tool.
- The system operates entirely in the terminal and provides immediate results upon completion of the quiz.

1.3 Features

The main features of the Online Quiz System are:

- Interactive Question-Answer Format: Presents a set of multiple-choice questions to the user in a clear and organized manner.
- Instant Evaluation: Checks each answer immediately and keeps track of the user's score.
- Case-Insensitive Input: Accepts both uppercase and lowercase answer choices (using toupper() function).User-Friendly Interface: Simple text-based interface suitable for beginners.

1. PROGRAM ANALYSIS

2.1 Problem Definition

The purpose of this project is to address the need for a simple and interactive system that allows users to participate in quizzes through a computer program. The following points define the problem:

1. Lack of interactive learning tools: There is a need for a lightweight, text-based quiz system that helps users test their knowledge without requiring complex software.
2. Limited use of basic C concepts: Beginners often struggle to find small, practical projects that apply fundamental C programming concepts in a fun way.
3. Manual quiz evaluation: Traditional quizzes require manual checking of answers; this program automates the evaluation process instantly.

4. User engagement and knowledge testing: There is a need for a simple digital system that combines entertainment and education by allowing users to attempt topic-based quizzes, such as food-related questions.

2.2 Project Overview

Aspect	Description
Language Used	C
Compiler	GCC / Dev-C++ / Turbo C++
Concepts Used	Conditional statements (if), loops (optional if expanded), character handling (toupper()), input/output operations
I/O Type	Character input (A/B/C/D options for answers)Text output displaying questions, score, and feedback
Optional	File Handling for saving receipts

2.3 Scopes and Limitations

Scope:

1.The project serves as an educational and entertainment tool that allows users to test their

general knowledge on specific topics such as food.2.It provides a base model for developing more advanced quiz applications, such as those with multiple topics, scoring systems, or graphical interfaces.

Limitations:

1. The quiz contains a fixed number of questions, which limits variety and replay value.
2. It does not store user scores or allow multiple users to participate simultaneously.
3. The system operates only in a text-based console environment, lacking graphical or online functionality.

1. SYSTEM DESIGN

3.1 Modules

The **Online Quiz System** is divided into simple, logical modules that work together to perform the complete quiz process. Each module handles a specific part of the system's functionality.

Module Name	Description
Input Module	This module is responsible for taking user inputs for each question in the quiz. It displays the multiple-choice options (A, B, C, D) and accepts the user's selected answer. The input character is processed using the scanf() function, and the toupper() function ensures that both uppercase and lowercase inputs are handled correctly.
Processing Module	The processing module performs the core logic of the quiz

	system. It compares the user's input with the correct answer using conditional statements (if conditions). Each correct answer increments the score counter (score++). Logical operations and decision-making form the basis of this module
Output Module	• This module displays the results and feedback to the user. • It prints each question, shows the final score at the end, and displays a thank-you message. • All outputs are generated using the printf() function in a clear and readable format.
File Handling	In the current version of the program, **file handling is not implemented**. However, in future enhancements, file handling can be used to: 1. Store user scores in a text file for record-keeping. 2. Load questions dynamically from a file instead of hardcoding them. 3. Maintain a leaderboard or user history for multiple quiz sessions

Result Display Module:	Calculates and displays the total score at the end of the quiz. Shows a thank-you message to conclude the session.

Module Interaction Overview

1.The Input Module collects the user's answers for each quiz question.2.The Processing Module evaluates the input, compares it with correct answers, and updates the score.3.The File Handling Module (if implemented) can store quiz data or retrieve previous records.4.Output Module displays messages, questions, and feedback during the quiz.5.The Result Display Module shows the final score and closing message to the user.

1. METHODOLOGY

The program is designed using basic C programming concepts.

It follows a step-by-step approach: display questions → take user input → check answers → calculate score → display result.

Functions like printf(), scanf(), and toupper() are used for input/output and answer checking.

4.1. Problem Analysis

Need for a simple quiz system to test user knowledge interactively.

Manual quiz checking is time-consuming; automation helps efficiency.

Users should get instant feedback on their performance.

The system should be easy to use and run in any C compiler.

4.2. System Design

1.Input Module: Takes user answers for each question.2.Processing Module: Checks answers and calculates the score.3.Output Module: Displays questions and final

results.4.File Handling (optional): Can be used to store scores or questions.

4.3. Program Development

1.Standard libraries like stdio.h and ctype.h are used for input/output and character handling.2.The program includes a sequence of multiple-choice questions, takes user input, checks answers, and displays the final score.3.The program is developed using C language in a simple and structured way.4.Basic programming constructs such as variables, if conditions, and loops (if needed) are used to build the quiz logic.

4.4. Testing and Validation

- The program was tested using different inputs (both uppercase and lowercase answers).
- Validation confirmed that correct answers increase the score and wrong answers do not.
- The program correctly displays the total score at the end.
- It runs successfully on GCC and Turbo C compilers without errors.

4.5. Result and Documentation

- **The program displays each question one by one and collects user responses.**
- **After all questions are answered, it shows the final score out of the total.**
- **A thank-you message appears, confirming the quiz completion.**
- **The output proves the quiz system works as expected and gives instant feedback to users.**
- **The project code is well-commented for easy understanding.**
- **Each step, from input to output, is explained using simple instructions.**
- **The documentation helps in understanding program flow, logic, and how modules interact.**

4.6. Algorithm

An **algorithm** is a **step-by-step logical procedure** used to solve a specific problem or perform a computation.In programming, it defines the exact sequence of actions the computer must follow to achieve the desired output from a given set of inputs.

For the ONLINE QUIZ SYSTEM, the algorithm plays a key role in automating the quiz

Purpose of the Algorithm

The main purpose of using an algorithm in this project is to:

- Define the **logical flow** of the answering the question.
- Ensure **accuracy** in calculations of score at the end .
- Provide a **structured path** for program development before actual coding.
- Make the system easy to understand, debug, and enhance in the future.

Algorithm in the Project

1.Start the program.

2.Display the quiz title and welcome message.

3.Initialize score = 0.

4.Display the first question with four options.

5.Take the user's input (A/B/C/D).

6.Convert input to uppercase using toupper().

7.Compare the input with the correct answer.

8.If correct, increase my score by 1.

9.Repeat for all questions.

10.Display the final score and thank-you message.

11. End the program

Importance of Algorithm in the Project

- The algorithm provides a clear structure for program flow.
- It helps in understanding how data moves between modules.
- Ensures the quiz runs logically and efficiently, producing correct results.
- Makes debugging and program improvement easier in the future.

4.7. Flowchart

Explanation of Flow

1. The program begins with a **start point**, where the quiz title and welcome message are displayed.

2. It initializes the **score** variable to zero to store the number of correct answers.
3. Each question is shown one by one with four multiple-choice options.
4. The **user enters an answer**, which is converted to uppercase to handle both lowercase and uppercase inputs.
5. The system **checks the correctness** of the answer using if conditions.
6. For every correct response, the score is **incremented by one**.
7. After all questions are answered, the program **displays the total score** and a **thank-you message** to the user.
8. Finally, the program reaches the **end** stage, marking the completion of the quiz.

```
│     Start      │
└──────────────┘
      │
┌──────────────────────────┐
│ Display Quiz Title │
│ and Welcome Msg    │
└──────────────────────────┘
      │
┌──────────────────────────┐
│ Initialize score=0 │
└──────────────────────────┘
      │
┌──────────────────────────┐
│ Display Question   │
│ and Options        │
└──────────────────────────┘
      │
```

```
┌──────────────────────────┐
│   Get User Input   │
│   (A/B/C/D)        │
└──────────┬───────────────┘
           │
┌──────────────────────────┐
│ Convert to Upper   │
│ using toupper()    │
└──────────┬───────────────┘
           │
┌──────────────────────────┐
│ Check Answer ==    │
│ Correct Option?    │
└──────────┬───────────────┘
           ┌──────────┐
           │          │
┌──────────▼──────┐   │
│Yes →    │       │
│score++  │       │
└──────────┬──────┘   │
           │          │
           │   No     │
           │          │
┌──────────▼──────────────┐
│ More Questions│
│ Remaining?    │
└──────────┬──────────────┘
           ┌──────────┐
```

```
      │         │
   ┌──────▼────┐   ┌──▼──────────┐
   │ Yes →  │   │ No     │
   │ Next Qn │   │ Display │
   │        │   │ Final   │
   │        │   │ Score   │
   └──────────┘   └──────────┘
      │         │
    ┌──▼──────────────────▼──┐
    │ Display Thank You      │
    │ Message                │
    └────────────────────────┘
          │
    ┌──────────────────────┐
    │      End             │
    └──────────────────────┘
```

1. SOURCE CODE

```
#include <stdio.h>

#include <ctype.h>


int main() {

    char ans;

    int score = 0;


    // Title + Welcome line
```

```
    printf("=== FOOD QUIZ ===\n");
    printf("Welcome to the fun world of food facts!\n\n");

    printf("1) King of Fruits?\nA) Apple B) Mango C) Orange D)
Banana\n");
    scanf(" %c", &ans);
    if (toupper(ans) == 'B') score++;

    printf("\n2) Country famous for Pizza?\nA) India B) USA C) Italy D)
China\n");
    scanf(" %c", &ans);
    if (toupper(ans) == 'C') score++;

    printf("\n3) Vitamin in citrus fruits?\nA) A B) B C) C D) D\n");
    scanf(" %c", &ans);
    if (toupper(ans) == 'C') score++;

    printf("\nYour Score: %d/3\n", score);
    printf("Thanks for playing!\n");

    return 0;
}
```

SOURCE CODE EXPLANATION:

Header Files

```
#include <stdio.h>
#include <ctype.h>
```

- **stdio.h** → This header file is used for standard input and output functions like printf() and scanf().
- **ctype.h** → It provides the function toupper(), which converts any lowercase character entered by the user into uppercase, making the program case-insensitive.

Main Function

```
int main() {
    char ans;
    int score = 0;
```

- The program execution starts from the main() function.
- A **character variable ans** is used to store the user's answer.
- An **integer variable score** is initialized to zero to keep track of correct answers.

Display Title and Welcome Message

```
printf("=== FOOD QUIZ ===\n");
printf("Welcome to the fun world of food facts!\n\n");
```

- Displays the quiz title and a welcome message on the screen to introduce the quiz.

Question 1

```
printf("1) King of Fruits?\nA) Apple B) Mango C) Orange D) Banana\n");
scanf(" %c", &ans);
if (toupper(ans) == 'B') score++;
```

- The first question is displayed with four multiple-choice options.

- The program waits for user input (A, B, C, or D).
- toupper(ans) ensures both ‘b’ and ‘B’ are treated the same.
- If the answer matches the correct option ('B' for Mango), the score is increased by 1.

Question 2

```
printf("\n2) Country famous for Pizza?\nA) India B) USA C) Italy D) China\n");
scanf(" %c", &ans);
if (toupper(ans) == 'C') score++;
```

- Displays the second question and takes the user’s response.
- If the user chooses 'C' (Italy), the score is incremented by 1.

Question 3

```
printf("\n3) Vitamin in citrus fruits?\nA) A B) B C) C D) D\n");
scanf(" %c", &ans);
if (toupper(ans) == 'C') score++;
```

- Displays the third question and checks the answer.
- If the user selects 'C' (Vitamin C), the score increases by 1.

Display Final Score

```
printf("\nYour Score: %d/3\n", score);
printf("Thanks for playing!\n");
```

- After all questions are answered, the total score (out of 3) is displayed.
- A thank-you message is shown to conclude the quiz.

Program End

```
return 0;
```

- Indicates successful program termination.

Summary

- The program uses **simple logic and conditional statements** to check answers.
- It demonstrates the use of **variables, input/output operations, conditional checks**, and **character handling** in C.
- The output is **interactive and user-friendly**, suitable for beginners learning programming fundamentals.

1. INPUT AND OUTPUT SCREENS

Input Screen 1:

```c
#include <stdio.h>
#include <ctype.h>

int main() {
    char ans;
    int score = 0;
    printf("=== FOOD QUIZ ===\n");

    printf("\n1) King of Fruits?\nA) Apple B) Mango C) Orange D) Banana\n");
    scanf(" %c",&ans);
    if(toupper(ans)=='B') score++;

    printf("\n2) Country famous for Pizza?\nA) India B) USA C) Italy D) China\n");
    scanf(" %c",&ans);
    if(toupper(ans)=='C') score++;

    printf("\n3) Vitamin in citrus fruits?\nA) A B) B C) C D) D\n");
    scanf(" %c",&ans);
    if(toupper(ans)=='C') score++;

    printf("\nYour Score: %d/3\n", score);
    return 0;
}
```

OUT PUT SCREEN

```
=== FOOD QUIZ ===

1) King of Fruits?
A) Apple B) Mango C) Orange D) Banana
Your answer: b

2) Country famous for Pizza?
A) India B) USA C) Italy D) China
Your answer: c

3) Vitamin in citrus fruits?
A) A B) B C) C D) D
Your answer: c

Your Score: 3/3
```

1. DISCUSSION & CONCLUSION

7.1 Discussion

The **Online Quiz System** was successfully developed using **C programming** to create an interactive question–answer game.

- The program works smoothly, displaying questions, taking user input, checking correctness, and calculating the final score automatically.
- The use of the toupper() function makes the program **case-insensitive**, improving

usability.

- It demonstrates how **basic programming concepts** like variables, conditions, and input/output operations can be combined to form a simple application.
- Although the program is text-based, it provides a foundation for future improvements, such as adding **file storage**, **more questions**, or a **graphical interface**.

7.2 Conclusion

- The project successfully meets its objective of creating a **simple and interactive quiz system** using C language.
- It helps users test their knowledge in a fun and educational way while showcasing fundamental programming logic.
- The quiz runs efficiently on any C compiler and produces accurate results.
- This project serves as a **beginner-level example** of applying basic coding concepts to solve real-world problems.
- Future enhancements can include adding Add GUI support, **integration with database**, and **extending functionality for scalability** to make the quiz more advanced and engaging.

1. REFERENCES

- Let Us C – Yashavant Kanetkar
- C Programming – E. Balagurusamy
- Online resources and documentation
- Dev-C++ User Manual